Dimension Theory of Metrizable Spaces and Games

C. Caruvana
with Steven Clontz

Indiana University Kokomo

Carolinas Topology Seminar
19 November 2021

Table of Contents

Notions of Topological Dimension

Countable-Dimensionality

Relative Covering Dimension

Games

Notions of Topological Dimension

Definition (Urysohn 1922/Menger 1923)

For a space X, we define the **small inductive dimension** ind as follows:

- $\operatorname{ind} X = -1$ if $X = \emptyset$.
- $\operatorname{ind} X \leqslant n$ ($n \in \omega$) if, for each $x \in X$ and open $U \subseteq X$ with $x \in U$, there exists an open $V \subseteq X$ so that $x \in V \subseteq \operatorname{cl}(V) \subseteq U$ and $\operatorname{ind} \partial V < n$.
- $\operatorname{ind} X = n$ if $\operatorname{ind} X \leqslant n$ and, for all $m < n$, $\neg \operatorname{ind} X \leqslant m$.
- $\operatorname{ind} X = \infty$ if, for all $n \in \omega$, $\neg \operatorname{ind} X \leqslant n$.

Note that, if there exists $n \in \omega$ so that $\operatorname{ind} X \leqslant n$, then X is regular.

Definition (L.E.J. Brouwer 1913/Čech 1931)

For a space X, we define the **large inductive dimension** Ind as follows:

- $\operatorname{Ind} X = -1$ if $X = \emptyset$.
- $\operatorname{Ind} X \leqslant n$ ($n \in \omega$) if, for each closed $F \subseteq X$ and open $U \subseteq X$ with $F \subseteq U$, there exists an open $V \subseteq X$ so that $F \subseteq V \subseteq \operatorname{cl}(V) \subseteq U$ and $\operatorname{Ind} \partial V < n$.
- $\operatorname{Ind} X = n$ if $\operatorname{Ind} X \leqslant n$ and, for all $m < n$, $\neg \operatorname{Ind} X \leqslant m$.
- $\operatorname{Ind} X = \infty$ if, for all $n \in \omega$, $\neg \operatorname{Ind} X \leqslant n$.

Note that, if there exists $n \in \omega$ so that $\operatorname{Ind} X \leqslant n$, then X is normal.

To define the covering dimension, we recall a few necessary tools.

Definition

For a family $\mathcal{A}$ of subsets of a set X that covers X and for $x \in X$, define $\mathrm{ord}_x(\mathcal{A}) = \#\{A \in \mathcal{A} : x \in A\}$, where $\#$ denotes cardinality. If $\mathcal{A}$ is so that $\mathrm{ord}_x(\mathcal{A}) < \omega$ for every $x \in X$, we define $\mathrm{ord}(\mathcal{A}) = \sup\{\mathrm{ord}_x(\mathcal{A}) - 1 : x \in X\}$.

Note that $\mathrm{ord}(\mathcal{A}) = 0$ if and only if $\mathcal{A}$ consists of pair-wise disjoint sets.

Definition

Let X be a space. For collections $\mathcal{A}$ and $\mathcal{B}$ of subsets of X, we say that $\mathcal{B}$ is an **open refinement** of $\mathcal{A}$ if $\mathcal{B}$ consists of open sets, $\mathcal{B}$ covers X, and, $(\forall B \in \mathcal{B})(\exists A \in \mathcal{A})\ B \subseteq A$.

Definition (Čech 1933/Lebesgue 1911)

For a space X, we define the **covering dimension** dim as follows:

- $\dim X \leqslant n$ ($n \in \omega$) if every open cover $\mathscr{U}$ of X admits an open refinement $\mathscr{V}$ so that $\operatorname{ord}(\mathscr{V}) \leqslant n$.
- $\dim X = n$ if $\dim X \leqslant n$ and, for all $m < n$, $\neg \dim X \leqslant m$.
- $\dim X = \infty$ if, for all $n \in \omega$, $\neg \dim X \leqslant n$.

Definition (Urbański 1995)

For a metrizable space X, we define **Urbański's transfinite dimension** d as follows:

- $dX = -1$ if $X = \emptyset$.
- $dX = 0$ if $\dim X = 0$.
- $dX \leqslant \alpha$ ($\alpha \in \omega_1$) if there exists a family $\{F_k : k \in \omega\}$ of closed subsets of X so that, for all $k \in \omega$, $dF_k < \alpha$ and $d\left(X \setminus \bigcup_{k\in\omega} F_k\right) \leqslant 0$.
- $dX = \infty$ if, for all countable ordinals α, $\neg dX \leqslant \alpha$.

As it turns out, for any metrizable space X and $n \in \omega$, $dX = n \iff \dim X = n$.

Theorem

For any separable metrizable space X and $n \in \omega$,

$$\operatorname{ind} X = n \iff \operatorname{Ind} X = n \iff \dim X = n \iff dX = n.$$

Theorem

For any separable metrizable space X and $n \in \omega$,

$$\operatorname{ind} X = n \iff \operatorname{Ind} X = n \iff \dim X = n \iff dX = n.$$

If we relax separability, we can still recover a portion of the previous result.

Theorem (Katětov 1952/Morita 1954)

For any metrizable space X, $\dim X = \operatorname{Ind} X$.

However, the small inductive dimension doesn't have to agree with the large inductive dimension, even in the realm of complete metrizable spaces.

Theorem (P. Roy 1962)

There exists a complete metrizable space X so that $\operatorname{ind} X = 0$ and $\operatorname{Ind} X = 1$.

Since $\operatorname{Ind} X = \dim X$ in the realm of metrizable spaces but $\operatorname{ind} X$ may disagree, and we'll be interested in zero-dimensional sets, we introduce the following notation.

Definition

We say that a space X is **zero-ind** (resp. **zero-dim**) if $\operatorname{ind} X = 0$ (resp. $\dim X = 0$).

Countable-Dimensionality

Definition (Hurewicz 1928)

A space is said to be **countable-dimensional** if it can be written as the countable union of finite-dimensional subspaces.

Countable-Dimensionality

Definition (Hurewicz 1928)

A space is said to be **countable-dimensional** if it can be written as the countable union of finite-dimensional subspaces.

In the realm of metrizable spaces, finite-dimensional subspaces can be written as the countable union of zero-dimensional subspaces. Hence, we can define countable-dimensional to be in terms of zero-dimensional subspaces.

Equipped with Roy's example, we would do well to distinguish between countable-dimensional relative to the small inductive dimension and the covering dimension.

Definition

A space is said to be σ**-zero-ind** if it can be written as the countable union of zero-ind subspaces. Analogously, a space is said to be σ**-zero-dim** if it can be written as the countable union of zero-dim subspaces

Since $\operatorname{ind} X \leqslant \dim X$ for metrizable X, we ask

Question

If a metrizable space is σ-zero-ind, is it also σ-zero-dim?

As we shall see, in the context of metrizable spaces, if a space is σ-zero-dim, it can be written as as the countable union of G_δ zero-dim subspaces. Can this be improved to closed subspaces?

Definition

A space is said to be **strongly countable-dimensional** if it can be written as the union of countably many closed finite-dimensional subspaces.

As we shall see, in the context of metrizable spaces, if a space is σ-zero-dim, it can be written as as the countable union of G_δ zero-dim subspaces. Can this be improved to closed subspaces?

Definition

A space is said to be **strongly countable-dimensional** if it can be written as the union of countably many closed finite-dimensional subspaces.

Observation

If X is strongly countable-dimensional, $dX \leqslant \omega$.

To see that countable-dimensional spaces aren't, in general, strongly countable-dimensional, we use the following.

Theorem

For a metrizable space X, if there is a countable ordinal α so that $dX \leqslant \alpha$, then X is countable-dimensional.

To see that countable-dimensional spaces aren't, in general, strongly countable-dimensional, we use the following.

Theorem

For a metrizable space X, if there is a countable ordinal α so that $dX \leqslant \alpha$, then X is countable-dimensional.

Proof.

We induct on α. Any finite-dimensional space is countable-dimensional, so we assume that, given some countable ordinal α, every metrizable space X satisfying $dX < \alpha$ is countable-dimensional. Suppose X is so that $dX = \alpha$ and write $X = G \cup \bigcup_{k \in \omega} F_k$ where, for each $k \in \omega$, F_k is closed and $dF_k < \alpha$ and $dG \leqslant 0$. Then, since F_k is countable-dimensional for each $k \in \omega$ by the inductive hypothesis, we see that X is countable-dimensional. $\square$

Theorem (Urbański 1995)

There exists a compact metrizable space X so that $dX = \omega + 1$.

This compact metrizable X with $dX = \omega + 1$ is countable-dimensional, but not strongly countable-dimensional.

Relative Covering Dimension

Definition (Valuyeva 1997)

For a space X and a subspace $Y \subseteq X$, we define the **relative covering dimension** of Y with respect to X as follows:

- $\dim_X Y \leqslant n$ ($n \in \omega$) if, for every collection $\mathscr{U}$ of open sets in X that covers Y, there exists a collection $\mathscr{V}$ of open sets in X that refines $\mathscr{U}$, covers Y, and $\operatorname{ord}(\mathscr{V}) \leqslant n$.
- $\dim_X Y = n$ if $\dim_X Y \leqslant n$ and, for all $m < n$, $\neg \dim_X Y \leqslant m$.
- $\dim_X Y = \infty$ if, for all $n \in \omega$, $\neg \dim_X Y \leqslant n$.

Definition

For a subspace $Y \subseteq X$, we say that Y is **zero-dim**$_X$ if $\dim_X Y = 0$. We say that X is σ**-zero-dim**$_X$ if X is the countable union of zero-dim$_X$ subspaces.

Proposition

For $Y \subseteq X$, $\dim Y \leqslant \dim_X Y$. Consequently, every zero-dim$_X$ subspace of X is zero-dim.

Proof.

Let $\mathscr{U}$ be an open cover of Y and, for each $U \in \mathscr{U}$, let $\hat{U}$ be open in X so that $\hat{U} \cap Y = U$. Then $\hat{\mathscr{U}} := \{\hat{U} : U \in \mathscr{U}\}$ is an open cover of Y by sets open in X so we can find a family $\mathscr{V}$ of sets open in X that refines $\hat{\mathscr{U}}$ and covers Y with $\operatorname{ord}(\mathscr{V}) \leqslant \dim_X Y$. Observe that $\mathscr{V}|_Y := \{V \cap Y : V \in \mathscr{V}\}$ is a refinement of $\mathscr{U}$ with $\operatorname{ord}(\mathscr{V}|_Y) \leqslant \dim_X Y$. $\square$

The previous inequality is, in general, strict since, under MA + ¬CH, there exists a normal space X with a subspace Y so that $\dim Y = 0$ but $\dim_X Y \neq 0$.

However,

Theorem

If X is metrizable and $Y \subseteq X$, then Y is zero-dim if and only if Y is zero-$\dim_X$.

Games

Definition

Let $\mathcal{A}$ and $\mathcal{B}$ be sets. We define the **single-selection game** $\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ as follows: At turn $n \in \omega$, One chooses $A_n \in \mathcal{A}$ and Two chooses $x_n \in A_n$. Two is declared the winner if $\{x_n : n \in \omega\} \in \mathcal{B}$. Otherwise, One wins.

Definition

For sets $\mathcal{A}$ and $\mathcal{B}$, the corresponding **finite-selection game** $\mathsf{G}_{\mathrm{fin}}(\mathcal{A}, \mathcal{B})$ is defined similarly, except that Two gets to choose $B_n \in [A_n]^{<\omega}$ at the nth stage and is declared the winner if $\bigcup_{n\in\omega} B_n \in \mathcal{B}$.

Many well-known games, like the Rothberger and Menger games and the point-open game, can be expressed as selection games.

Notation

Given a space X, we will let $\mathscr{T}_X$ denote the set of all non-empty, proper open subsets of X and $\mathcal{O}_X$ denote the set of all non-trivial open covers of X; that is, all open covers $\mathscr{U}$ of X so that $X \notin \mathscr{U}$. When the context is clear, we will simply write $\mathcal{O}$ to denote $\mathcal{O}_X$.

The following game is inspired by Bing's screenability condition and Addis and Gresham's selective version of screenability.

Definition

Given a space X, the game $\mathsf{G}_c(\mathcal{O}, \mathcal{O})$ is defined as follows: At turn $n \in \omega$, One chooses an open cover $\mathscr{U}_n$ of X and Two chooses a pairwise disjoint family $\mathscr{V}_n$ of open sets that refines $\mathscr{U}_n$. Two is declared the winner if $\bigcup_{n\in\omega} \bigcup \mathscr{V}_n = X$. Otherwise, One wins.

To illustrate the generality of our selection game framework, we could cast $\mathsf{G}_c(\mathcal{O}, \mathcal{O})$ as a single-selection game if we define, for an open cover $\mathscr{U}$ of X, $\mathrm{pdr}(\mathscr{U})$ to be the set of all pairwise disjoint families of open sets that refine $\mathscr{U}$,
$\mathcal{A} = \{\mathrm{pdr}(\mathscr{U}) : \mathscr{U} \in \mathcal{O}\}$, and

$$\mathcal{O}^* = \left\{ \mathscr{V} \in \wp(\mathscr{T}_X)^\omega : \bigcup_{n\in\omega} \bigcup \mathscr{V}_n = X \right\}.$$

Then $\mathsf{G}_1(\mathcal{A}, \mathcal{O}^*)$ is precisely the game $\mathsf{G}_c(\mathcal{O}, \mathcal{O})$.

Another observation is the following: note that, given an open cover $\mathscr{U}$ of X,

$$\hat{\mathscr{U}} := \left\{ \bigcup \mathscr{V} : \mathscr{V} \in \mathrm{pdr}(\mathscr{U}) \right\}$$

is an open cover of X so that, for every zero-dim$_X$ subspace Y, there exists $U \in \hat{\mathscr{U}}$ so that $Y \subseteq U$.

Another observation is the following: note that, given an open cover $\mathscr{U}$ of X,

$$\hat{\mathscr{U}} := \left\{ \bigcup \mathscr{V} : \mathscr{V} \in \mathrm{pdr}(\mathscr{U}) \right\}$$

is an open cover of X so that, for every zero-dim$_X$ subspace Y, there exists $U \in \hat{\mathscr{U}}$ so that $Y \subseteq U$.

Moreover, if Two pretends like they're choosing from $\hat{\mathscr{U}}$ instead of picking pairwise disjoint refinements of $\mathscr{U}$, their winning condition doesn't change.

Definition

Let $\mathcal{O}_{\mathrm{zd}}$ be the set of all non-trivial open covers $\mathscr{U}$ of X so that, for each zero-dim$_X$ subspace Y, there exists $U \in \mathscr{U}$ so that $Y \subseteq U$.

How is it that $\mathsf{G}_c(\mathcal{O}, \mathcal{O})$ and $\mathsf{G}_1(\mathcal{O}_{\mathrm{zd}}, \mathcal{O})$ compare?

We consider four main types of strategies for games.

- A **predetermined winning strategy for One** relies only on the turn number. That is, it tells One how to win the game by playing a particular sequence.
- A **perfect information winning strategy for One** relies on Two's responses along the way; it tells One how to respond at round n given Two's responses in the first $n-1$ rounds so that One wins the run of the game.

- A **perfect information winning strategy for Two** relies on One's plays along the way; it tells Two how to respond at round n given One's responses in the first $n-1$ so that Two wins the run of the game.
- A **Markov winning strategy for Two** relies only on the turn number and One's most recent play; that is, Two only needs to know One's play A_n at round n to respond in such a way that Two wins the run of the game.

- A **perfect information winning strategy for Two** relies on One's plays along the way; it tells Two how to respond at round n given One's responses in the first $n-1$ so that Two wins the run of the game.
- A **Markov winning strategy for Two** relies only on the turn number and One's most recent play; that is, Two only needs to know One's play A_n at round n to respond in such a way that Two wins the run of the game.

Observation

Traditional selection principles are equivalent to One not having a predetermined winning strategy in the corresponding selection game.

Theorem

Two has a Markov winning strategy in $\mathsf{G}_c(\mathcal{O}, \mathcal{O})$ if and only if X is σ-zero-dim$_X$.

Proof.

$\Rightarrow$: Let τ represent Two's Markov winning strategy. Define

$$Y_n = \bigcap_{\mathscr{U} \in \mathcal{O}} \bigcup \tau(\mathscr{U}, n)$$

and notice that Y_n is zero-dim$_X$. If $x \notin Y_n$ for all $n \in \omega$, then we could choose $\mathscr{U}_n \in \mathcal{O}$ so that $x \notin \bigcup \tau(\mathscr{U}_n, n)$. But then the sequence $\langle \mathscr{U}_n : n \in \omega \rangle$ is a play by One that defeats Two, contradicting the fact that τ is winning.

$\Leftarrow$: Suppose $X = \bigcup_{n \in \omega} Y_n$ where Y_n is zero-dim$_X$. Then, at round n, given an open cover $\mathscr{U}_n$ of X, let Two respond with a pairwise disjoint refinement of $\mathscr{U}_n$ that covers Y_n. $\square$

A similar proof establishes that

Theorem

Two has a Markov winning strategy in $\mathsf{G}_1(\mathcal{O}_{\mathrm{zd}}, \mathcal{O})$ if and only if X is σ-zero-dim$_X$.

To establish results for perfect information strategies in $\mathsf{G}_1(\mathcal{O}_{\mathrm{zd}}, \mathcal{O})$, we start by establishing a connection with the "point-picking" game.

Notation

For a subspace Y of X, let $\mathcal{N}(Y) = \{U \in \mathscr{T}_X : Y \subseteq U\}$ and $\mathcal{N}_{\mathrm{zd}} = \{\mathcal{N}(Y) : Y \subseteq X \text{ is zero-dim}_X\}$.

Then the game $\mathsf{G}_1(\mathcal{N}_{\mathrm{zd}}, \neg\mathcal{O})$ is equivalent to the game where, in round n, One picks a zero-dim$_X$ subspace Y_n and Two responds with an open set U_n that contains Y_n with the winning condition for Two being that $\{U_n : n \in \omega\}$ doesn't cover the space.

The following is analogous to the well-known duality (Galvin 1978) of the point-open game and the Rothberger game and the proof follows from general tools in Clontz 2020.

Theorem

The games $\mathfrak{G} = \mathsf{G}_1(\mathcal{O}_{\mathrm{zd}}, \mathcal{O})$ and $\mathfrak{H} = \mathsf{G}_1(\mathcal{N}_{\mathrm{zd}}, \neg\mathcal{O})$ are dual; that is,

- One has a predetermined winning strategy for $\mathfrak{G}$ iff Two has a Markov winning strategy for $\mathfrak{H}$.
- One has a perfect information winning strategy for $\mathfrak{G}$ iff Two has a perfect information winning strategy for $\mathfrak{H}$.
- Two has a perfect information winning strategy for $\mathfrak{G}$ iff One has a perfect information winning strategy for $\mathfrak{H}$.
- Two has a Markov winning strategy for $\mathfrak{G}$ iff One has a predetermined winning strategy for $\mathfrak{H}$.

Another important ingredient:

Theorem

If X is metrizable and $Y \subseteq X$ is so that $\operatorname{Ind} Y = 0$, then there exists a G_δ set G so that $Y \subseteq G$ and $\operatorname{Ind} G = 0$.

By replacing all zero-dim$_X$ subspaces with zero-dim$_X$ G_δ sets and following Telgársky 1975, we obtain

Theorem

For a metrizable space X, the following are equivalent:

- X is σ-zero-dim.
- One has a predetermined winning strategy in $\mathsf{G}_1(\mathcal{N}_{\mathrm{zd}}, \neg\mathcal{O})$.
- One has a perfect information winning strategy in $\mathsf{G}_1(\mathcal{N}_{\mathrm{zd}}, \neg\mathcal{O})$.

Hence, by duality,

Corollary

For a metrizable space X, the following are equivalent:

- X is σ-zero-dim.
- Two has a Markov winning strategy in $\mathsf{G}_1(\mathcal{O}_{\mathrm{zd}}, \mathcal{O})$.
- Two has a perfect information winning strategy in $\mathsf{G}_1(\mathcal{O}_{\mathrm{zd}}, \mathcal{O})$.
- Two has a Markov winning strategy in $\mathsf{G}_c(\mathcal{O}, \mathcal{O})$.

Hence, by duality,

Corollary

For a metrizable space X, the following are equivalent:

- X is σ-zero-dim.
- Two has a Markov winning strategy in $\mathsf{G}_1(\mathcal{O}_{\mathrm{zd}}, \mathcal{O})$.
- Two has a perfect information winning strategy in $\mathsf{G}_1(\mathcal{O}_{\mathrm{zd}}, \mathcal{O})$.
- Two has a Markov winning strategy in $\mathsf{G}_c(\mathcal{O}, \mathcal{O})$.

Question

For all metrizable X, can we add the condition that Two has a perfect information winning strategy in $\mathsf{G}_c(\mathcal{O}, \mathcal{O})$ into the equivalence above?

In the context of metrizable spaces, we see that, in $\mathsf{G}_1(\mathcal{N}_{\mathrm{zd}}, \neg\mathcal{O})$, One can play G_δ sets. However, One cannot improve to closed sets, in general, even if we allow One to play closed finite-dimensional sets. To demonstrate this, we first note a generalization of Telgársky's result that the point-open and finite-open games are equivalent.

Theorem

If $\mathcal{N}_{\mathrm{fd}}$ is the collection of $\mathcal{N}(Y)$ where Y is a finite-dimensional subspace of a separable metrizable space X, then $\mathsf{G}_1(\mathcal{N}_{\mathrm{zd}}, \neg\mathcal{O})$ is equivalent to $\mathsf{G}_1(\mathcal{N}_{\mathrm{fd}}, \neg\mathcal{O})$.

Note that this uses the fact that, for separable metrizable X, finite-dimensional subspaces can be written as the finite union of zero-dimensional subspaces.

Now, if $\mathcal{N}_{\mathrm{cfd}}$ is the collection of $\mathcal{N}(Y)$ where Y is a closed finite-dimensional subspace of a metrizable space X, we have that

Theorem

The following are equivalent:

- X is strongly countable-dimensional.
- One has a predetermined winning strategy in $\mathsf{G}_1(\mathcal{N}_{\mathrm{cfd}}, \neg\mathcal{O})$.
- One has a perfect information winning strategy in $\mathsf{G}_1(\mathcal{N}_{\mathrm{cfd}}, \neg\mathcal{O})$.

Note that Urbański's example of a compact metric space X with $dX = \omega + 1$ is countable-dimensional, but not strongly countable-dimensional. Hence, X is a space where One has a predetermined winning strategy in $\mathsf{G}_1(\mathcal{N}_{\mathrm{zd}}, \neg\mathcal{O})$, but fails to have even a perfect information winning strategy in $\mathsf{G}_1(\mathcal{N}_{\mathrm{cfd}}, \neg\mathcal{O})$.

By duality, we have that $\mathsf{G}_1(\mathcal{O}_{\mathrm{zd}}, \mathcal{O})$ and $\mathsf{G}_1(\mathcal{O}_{\mathrm{fd}}, \mathcal{O})$ are equivalent for separable metrizable X, were $\mathcal{O}_{\mathrm{fd}}$ is the collection of all open covers so that every finite-dimensional subspace is contained in some element of the cover.

Theorem (Babinkostova 2007)

By an example of E. Pol (1993), there exists a separable metrizable space X where One has a predetermined winning strategy in $\mathsf{G}_1(\mathcal{O}_{\mathrm{fd}}, \mathcal{O})$ (and, hence, for $\mathsf{G}_1(\mathcal{O}_{\mathrm{zd}}, \mathcal{O})$), but One does not have a predetermined winning strategy in $\mathsf{G}_c(\mathcal{O}, \mathcal{O})$.

From this, it is apparent that $\mathsf{G}_c(\mathcal{O}, \mathcal{O})$ and $\mathsf{G}_1(\mathcal{O}_{\mathrm{zd}}, \mathcal{O})$ are not equivalent, despite their similarity on Two's side of things.

Thank You!